TITANIC SURVIVAL ANALYSIS

Excel Data Analysis Project Report

Introduction

The Titanic disaster remains one of history's most widely remembered maritime tragedies. Passenger records also provide an opportunity to examine survival patterns and understand how outcomes differed across passenger groups. This project uses Microsoft Excel to turn those records into clear summaries and visual comparisons.

The goal is to present the available data in a practical, accessible way. By combining key measures with charts, the dashboard makes it easier to see the overall outcome and explore differences by gender, passenger class, age group, and fare range.

1. Project Objectives

This project was designed to answer five questions: How many passenger records are included? How many passengers are recorded as survivors, and what proportion does that represent? How do outcomes compare across gender and passenger class? What patterns appear across age groups and fare ranges? How can Excel communicate these findings clearly through a dashboard?

2. Dataset Overview

The dashboard summarizes 1,309 passenger records. The workbook is organized into four sheets: Transformed Dataset, Dataset Dictionary, Worksheet, and Dashboard. The raw dataset is kept separate from the transformed dataset, providing a clear structure for the analysis.

Passenger fields used in this type of analysis include characteristics such as gender, age, passenger class, fare, and survival status. These fields make it possible to summarize outcomes and compare groups. The Dataset Dictionary provides the field definitions used in the workbook.

3. Analytical Approach

Microsoft Excel was used to organize the data, calculate summary measures, and present the results visually.

The dashboard combines KPI cards with charts covering passenger class, gender, age group, and fare range.
This layout gives readers a quick overview and makes specific comparisons easier to follow.

The analysis follows a clear sequence: organize the available records, prepare a transformed dataset, summarize the main measures, and communicate the results through a dashboard. Keeping the raw and transformed datasets separate also makes the preparation process easier to review.

4. Key Dashboard Results

Measure	Dashboard result
Total passengers	1,309
Total survivors	500
Female survivors	339
Overall survivor rate	38.20%
Average fare per passenger	$33.28

The dashboard presents five headline measures that summarize the dataset and provide the foundation for the comparisons that follow.

5. Findings and Interpretation

The dataset contains 1,309 passenger records, of which 500 are recorded as survivors. This produces an overall survival rate of 38.20%. The result gives a clear overview of the outcome represented in the dataset and provides a baseline for examining differences between passenger groups.

The dashboard records 339 female survivors. This highlights the number of women recorded as surviving, while the gender chart provides a basis for examining how outcomes differed by gender. Comparing the likelihood of survival between groups requires considering survivor counts alongside the total number of passengers in each group.

The passenger-class chart supports comparison of survival outcomes across classes. The age-group and fare-range charts add further

perspectives by showing how outcomes are distributed across those categories.
Together, these views make the dashboard more informative than a single overall statistic.

The average fare per passenger is $33.28. Alongside the survival measures, this provides a useful summary of the fare values in the dataset and adds context to the fare-range analysis.

6. Value of the Dashboard

The dashboard brings the main results together in one place, reducing the need to inspect individual records to understand the broad picture. KPI cards communicate headline figures immediately, while charts make group comparisons easier to follow.

This project demonstrates how Excel can support data analysis and reporting: a dataset becomes a structured set of measures and visual summaries that can be reviewed and communicated clearly.

7. Conclusion

This project demonstrates the use of Microsoft Excel to organize passenger data, summarize survival outcomes, and communicate findings through a focused dashboard. The dashboard reports 1,309 passenger records, 500 survivors, an overall survival rate of 38.20%, 339 female survivors, and an average fare of $33.28.

By bringing these measures together with comparisons by passenger class, gender, age group, and fare range, the project provides a clear overview of the available data. It demonstrates core analytical skills in data organization, summary reporting, and dashboard presentation.

8. Project Deliverables

The project package consists of the raw dataset, transformed dataset, Excel workbook, dashboard screenshot, and this report. Together, these deliverables present the source data, prepared data, analysis workbook, and dashboard findings in a structured portfolio project.